SQLite - Data Type

SQLite data type is an attribute that specifies the type of data of any object. Each column, variable and expression has related data type in SQLite.

You would use these data types while creating your tables. SQLite uses a more general dynamic type system. In SQLite, the datatype of a value is associated with the value itself, not with its container.

SQLite Storage Classes

Each value stored in an SQLite database has one of the following storage classes −

Sr.No.	Storage Class & Description
1	**NULL** The value is a NULL value.
2	**INTEGER** The value is a signed integer, stored in 1, 2, 3, 4, 6, or 8 bytes depending on the magnitude of the value.
3	**REAL** The value is a floating point value, stored as an 8-byte IEEE floating point number.
4	**TEXT** The value is a text string, stored using the database encoding (UTF-8, UTF-16BE or UTF-16LE)
5	**BLOB** The value is a blob of data, stored exactly as it was input.

SQLite storage class is slightly more general than a datatype. The INTEGER storage class, for example, includes 6 different integer datatypes of different lengths.

SQLite Affinity Type

SQLite supports the concept of **type affinity** on columns. Any column can still store any type of data but the preferred storage class for a column is called its **affinity**. Each table column in an SQLite3 database is assigned one of the following type affinities –

Sr.No.	Affinity & Description
1	**TEXT** This column stores all data using storage classes NULL, TEXT or BLOB.
2	**NUMERIC** This column may contain values using all five storage classes.
3	**INTEGER** Behaves the same as a column with NUMERIC affinity, with an exception in a CAST expression.
4	**REAL** Behaves like a column with NUMERIC affinity except that it forces integer values into floating point representation.
5	**NONE** A column with affinity NONE does not prefer one storage class over another and no attempt is made to coerce data from one storage class into another.

SQLite Affinity and Type Names

Following table lists down various data type names which can be used while creating SQLite3 tables with the corresponding applied affinity.

Data Type	Affinity
• INT • INTEGER • TINYINT • SMALLINT	INTEGER

• MEDIUMINT • BIGINT • UNSIGNED BIG INT • INT2 • INT8	
• CHARACTER(20) • VARCHAR(255) • VARYING CHARACTER(255) • NCHAR(55) • NATIVE CHARACTER(70) • NVARCHAR(100) • TEXT • CLOB	TEXT
• BLOB • no datatype specified	NONE
• REAL • DOUBLE • DOUBLE PRECISION • FLOAT	REAL
• NUMERIC • DECIMAL(10,5) • BOOLEAN • DATE • DATETIME	NUMERIC

Boolean Datatype

SQLite does not have a separate Boolean storage class. Instead, Boolean values are stored as integers 0 (false) and 1 (true).

Date and Time Datatype

SQLite does not have a separate storage class for storing dates and/or times, but SQLite is capable of storing dates and times as TEXT, REAL or INTEGER values.

Sr.No.	Storage Class & Date Formate
1	**TEXT** A date in a format like "YYYY-MM-DD HH:MM:SS.SSS"
2	**REAL** The number of days since noon in Greenwich on November 24, 4714 B.C.
3	**INTEGER** The number of seconds since 1970-01-01 00:00:00 UTC

You can choose to store dates and times in any of these formats and freely convert between formats using the built-in date and time functions.